



University of Vavuniya

Second Examination in Information Technology – 2022

First Semester – January/February 2024

IT2133 – Advanced Web Programming (Theory)

Answer any Four Questions

Time Allowed : Two hours

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1. (a) List the five basic elements of web design. [10%]
(b) Briefly describe four ways to reduce the page loading time. [20%]
(c) Describe the significant differences between *static* and *dynamic* web pages and list down sample web contents for each of the types. [20%]
(d) Briefly explain why PHP is the best choice for web application development. [20%]
(e) List down any *five* features of the Domain Name System(DNS). [10%]
(f) "JavaScript supports automatic type conversion" Justify this statement. [20%]
2. (a) What is the meaning of a security threat to a system and provide an example? [15%]
(b) Briefly describe *four* key elements of computer security. [20%]
(c) Write short notes on each of the following types of cyber-attacks:
 - i. SQL injection
 - ii. Pharming
 - iii. Denial of service
 - iv. DNS Spoofing [20%]

[This question is continued on the next page]

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✱ (d) Describe the *four* consequences of ignoring the importance of computer security. [20%]

(e) Discuss the best practices required to follow by users to prevent a system from cyber-attacks. [25%]

backup/
Antivirus
logics
update

3. (a) Discuss how JavaScript is different from Java. [15%]

(b) Briefly describe each of the following types of *Error Name Values* in JavaScript.

i. Range Error

ii. Reference Error

iii. Syntax Error

iv. Type Error

[20%]

(c) State two different ways of embedding JavaScript code in an HTML file. [15%]

(d) Explain the importance of jQuery in JavaScript. [15%]

JS is simple & easy to use

(e) Briefly explain the Document Object Model (DOM) tree structure of HTML elements with a suitable example. [20%]

(f) List the steps for creating HTML elements using JavaScript with examples. [15%]

4. (a) In jQuery *.each* construct can be used to iterate through a collection of jQuery objects. Briefly describe the syntax and semantics of *.each* construct. [20%]

(b) Identify the object, property and method in the following JavaScript code segment and add a new property to the given object:

```
var student = { name: "John", marks: 27, getdata: function ()  
{ return this.name + " " + this.marks; } };
```

[15%]

(c) Write the jQuery syntax to select all DOM elements with the id="id01" or class "option". [10%]

(d) State the purpose of jQuery *toggleClass* function. [15%]

[This question is continued on the next page]

- (e) Write a script that uses only jQuery to find an id of the following HTML element.

```
<div id="id01" name="test" >
    This is a division
</div>
```

[20%]

- (f) Write the syntax for the following jQuery selectors:

- i. Tag name
- ii. ID
- iii. Class
- iv. Attribute
- v. Attribute value

[20%]

5. (a) Write the output for the given code segment.

```
<?php
echo "10" > 1 , "\n";
echo "10 items" > 2 , "\n";
echo "items" > 2 , "\n";
echo True == 1, "\n";
echo TRUE == FALSE, "\n";
?>
```

[10%]

- (b) State the significant difference between *echo* and *print* commands in PHP. [15%]
- (c) Describe what is meant by HTTP cookies and list down its components. Describe the advantages of web cookies. [25%]
- (d) State the significant difference between functions *mysqli_fetchobject()* and *mysqli_fetch_array()*. [15%]
- (e) Discuss the differences between functions *require()* and *include()*. [20%]
- (f) Compare and contrast "GET" and "POST" methods in PHP. [15%]